

Locomotion-Grounded Soccer Skills: A Unified Humanoid Policy for Free Navigation and Multi Motion-Guided Kicking Skills

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1 Problem Description

Task: Learn a *single* low-level policy for a 29-DoF humanoid that simultaneously (i) accepts arbitrary locomotion commands (v_x, v_y, ω_z) over varied terrain, and (ii) executes N distinct motion-clip-derived kicking skills, with smooth bidirectional transitions between the locomotion mode and each skill.

Problem: Recent humanoid soccer systems obtain whole-body stability by making *motion tracking the substrate* and deriving locomotion from it, typically by steering a motion-reference anchor toward the ball [Zhong et al., 2026, Kong et al., 2026, Wang et al., 2026]. This couples two limitations. First, because tracking rewards remain active throughout the approach, the reachable gait space is confined to a neighbourhood of the reference clip’s own locomotion content, so behaviours absent from that clip—sustained lateral walking, backpedalling, spin-in-place, running—are neither trained nor evaluated. Second, these systems rely on a *single* reference kick, which bounds the achievable strike repertoire: Zhong et al. [2026] report a “natural right-footed bias” and degradation concentrated “at extreme lateral and high targets”, and name multi-skill priors as their principal open problem. A real match, however, requires free navigation (repositioning, evading opponents, retreating) interleaved with a *library* of strike types (inside-foot pass, instep drive, lateral flick, weak-foot strike).

Novelty: We invert the architecture. A general, commandable locomotion policy is the substrate; N motion-tracked kicking skills are task-gated layers on top of it. Concretely:

- **Substrate inversion.** Skills are built *on* locomotion rather than locomotion being built *from* tracking, decoupling the reachable gait space from any reference clip.
- **Locomotion as a composition hub.** Every skill initiates from and terminates into the same commandable locomotion state, so a library of N skills requires $O(N)$ transition specifications rather than $O(N^2)$ pairwise ones.
- **Learned handoff instead of a scripted stabilisation tail.** At the end of authored reference content, control is returned to the locomotion policy rather than tracking a synthetic recovery clip.
- **A multi-skill strike library** that extends target-direction and ball-placement coverage beyond what a single forward reference can express, including passing, whose contact mechanics differ in kind—not merely in target location—from shooting.

We model the problem as a task-conditioned MDP. A permanent per-environment task assignment selects locomotion or one of N skills; a task-mode variable $m_t \in \{\text{LOCO}, \text{KICK}_1, \dots, \text{KICK}_N\}$ gates both the observation (via a task/skill one-hot) and every reward and termination term. In locomotion mode the policy tracks (v_x, v_y, ω_z) ; in skill mode it tracks skill i ’s retargeted human reference while the ball/target rewards shape contact and aim. A skill terminates when its authored reference content is exhausted, at which point m_t returns to LOCO and the locomotion policy resumes control.

2 Background

2.1 Humanoid Soccer and the Shooting Problem

RL has produced athletic sports behaviours for quadruped soccer, table tennis and badminton [Su et al., 2025a,b, Ma et al., 2025], and a rapidly growing line of humanoid soccer systems [Haarnoja et al., 2024, Wang et al., 2025, Xu et al., 2026, Wang et al., 2026, Kong et al., 2026, Li et al., 2025, Ren et al., 2025]. Haarnoja et al. [2024] demonstrated end-to-end multi-skill soccer (walk, turn, kick, get-up) with learned transitions on a miniature bipedal platform. On full-size humanoids, Zhong et al. [2026] report the strongest shooting results to date—0.73 m mean target error from 3 m and peak ball speeds of 13.10 m/s on a Unitree G1—via a three-stage motion-guided curriculum.

2.2 Motion Tracking and Imitation for Whole-Body Control

Reference-guided RL provides the whole-body coordination that pure task rewards struggle to discover [Peng et al., 2018, 2021]. Recent humanoid systems track retargeted human motion directly [Liao et al., 2025], using video-based recovery [Shen et al., 2024] and general motion retargeting [Araújo et al., 2025] to obtain references. Tracking supplies stability but constrains timing and contact selection, which is precisely the tension staged curricula are designed to resolve [Zhong et al., 2026].

2.3 Commandable Locomotion Policies

Massively parallel simulation has made robust, velocity-commanded legged locomotion with terrain and disturbance curricula a well-established primitive [Rudin et al., 2022, Mittal et al., 2023]. This line of work yields policies that track arbitrary commands over rough terrain—the capability our architecture places at the *base* of the stack rather than deriving as a by-product of tracking.

2.4 Whole-Body Object Interaction

Contact-rich loco-manipulation typically assumes dwell-time contact and accumulates interaction reward over many steps [Weng et al., 2025, Zhao et al., 2025, Xu et al., 2025, Zhang et al., 2026]. Shooting instead involves a sub-10 ms impulse with delayed ball-target feedback, motivating specialised interaction rewards [Zhong et al., 2026].

2.5 Skill Composition and Transitions

The options framework formalises temporally extended actions through initiation sets, policies and termination conditions [Sutton et al., 1999]. A skill library is composable when each skill’s termination set lies within the next skill’s initiation set; our locomotion hub makes this condition hold by construction, since every skill terminates into the same commandable locomotion state.

3 Limitation and Novelty

Limitations of Prior Work

- **Gait space bounded by the reference clip.** In Zhong et al. [2026], motion-tracking rewards are discounted but never disabled during approach; locomotion is realised by steering the tracking anchor toward a predicted ball position. Gaits with no counterpart in the reference are therefore not reachable, independent of the command issued.

- **Locomotion generality untested.** All reported experiments in [Zhong et al., 2026, Kong et al., 2026, Xu et al., 2026] are free-kick or moving-ball shooting. No locomotion metrics—command tracking, terrain, push recovery—are reported, so the command distribution outside ball approach is unevaluated.
- **Single-skill repertoire.** Zhong et al. [2026] rely on one right-footed side-foot reference and report a right-footed bias and degradation at extreme lateral and high targets; their Limitations section names multi-skill priors and learned skill selection as future work.
- **Scripted post-skill stabilisation.** After contact, a synthetic stabilisation reference is held for T_{stab} steps. This component is load-bearing—removing it drops alive rate from 98.8% to 24.4%—but it is momentum-blind, authored per clip, and does not generalise across a skill library.
- **Handcrafted skill triggering.** Stage transitions and kick timing are governed by a rule-based planner, which Zhong et al. [2026] identify as limiting autonomous decision making.

Table 1 summarises these gaps across the humanoid soccer literature. Two columns are empty for every prior system: no reported evaluation of locomotion as a general capability, and no strike repertoire extending beyond a single dominant-foot contact type. These are the two axes our architecture is designed to occupy.

Method	Platform	Locomotion substrate		Skill repertoire & transitions				
		Free cmd.	Omni dir.	Multi skill	Motion guided	Lateral tgt.	Loco→Kick	Kick→Loco
Haarnoja et al. [Haarnoja et al., 2024]	Miniature	X [†]	○	X [‡]	X	○	✓	✓
STOFT [Li et al., 2025]	Bipedal	X	X	X	X	X	X	X
Reactive [Wang et al., 2025]	Humanoid	○	X	○	✓	X	X	X
Striker [Xu et al., 2026]	Humanoid	○	X	X	X	○	X	X
Goalkeeper [Ren et al., 2025]	Humanoid	X	X	X	✓	X	X	X
HumanX [Wang et al., 2026]	Humanoid	X	X	✓	✓	X	X	X
PAiD [Kong et al., 2026]	Humanoid	○	X	X	✓	○	X	X
RoboNaldo [Zhong et al., 2026]	Humanoid	○	X	X	✓	○	X	○
Ours (proposed)	Humanoid	✓	✓	✓	✓	✓	✓	✓

Table 1: **Capability comparison against prior humanoid soccer systems.** ✓ = supported and demonstrated; ○ = mechanism present but restricted to the ball-approach regime, or not evaluated as a general capability; X = absent. *Free cmd.*: policy accepts arbitrary externally issued velocity commands rather than commands emitted solely by a ball-approach planner. *Omni dir.*: sustained lateral, backward and in-place rotation demonstrated. *Multi skill*: more than one distinct strike/contact type. *Motion guided*: the skill’s reward is shaped by tracking or matching human reference motion (frame-exact clip tracking or adversarial motion-prior matching), rather than emerging purely from a task/outcome reward or a non-learned trajectory. *Lateral tgt.*: demonstrated strike accuracy to off-axis, non-forward-cone targets, not just a narrow forward target. *Loco→Kick*: entry into a kicking skill is a handoff from a trained locomotion controller, rather than a kick-tracking reward that stays active throughout approach with no separate locomotion-mode substrate to hand off from. *Kick→Loco*: post-strike control returns to a trained locomotion controller rather than a scripted stabilisation reference. Marks reflect capabilities *as reported in each paper*; an unreported capability is scored X or ○ rather than assumed absent. The ○ in RoboNaldo’s *Free cmd.* column reflects a locomotion-command interface whose commands originate from a ball-approach heuristic, with tracking rewards active throughout; its ○ under *Lateral tgt.* reflects reported degradation at extreme lateral and high targets rather than uniform coverage; its X under *Loco→Kick* reflects that the same always-active tracking reward leaves no separate locomotion-mode substrate to hand off from; and its ○ under *Kick→Loco* reflects a scripted per-clip stabilisation reference rather than a return to a trained locomotion controller. [†]Haarnoja et al.’s policy receives full egocentric game state (ball/opponent/goal position and velocity) rather than an externally issued velocity command, and has no command interface for a human or high-level planner to direct locomotion independent of the game – it is autonomous game-state-conditioned behaviour, not free command tracking as defined above. [‡]Their agent exhibits a behaviour repertoire (walk, turn, kick, get up) but a single scoring strike learned against a stationary opponent, with no second contact type reported; the *Multi skill* column tracks strike/contact-type diversity specifically, not behavioural diversity.

Novelty

- **Locomotion-first architecture:** a commandable locomotion policy as substrate, with task-gated motion-tracked skills layered on top, yielding a reachable gait space independent of any clip.
- $O(N)$ **skill composition** via a locomotion hub state shared by all skills.
- **Learned-controller handoff** replacing scripted post-skill stabilisation, motivated by failure telemetry localising the dominant instability to this window.
- **Multi-skill strike library** broadening target-direction and ball-placement coverage, and enabling passing as a distinct contact modality.

- **Joint evaluation** of locomotion generality *and* shooting quality in one policy—a combination not reported by prior humanoid soccer systems.

4 Description of Database/Performance Measure/Current SOTA

4.1 Platform, Data and Environments

- **Robot:** Unitree G1, 29 DoF. Policy at 50 Hz, deployed as ONNX with onboard inference.
- **Motion data:** N human kicking clips (inside-foot pass, instep drive, lateral/flick, weak-foot strike) recovered from video [Shen et al., 2024] and retargeted [Araújo et al., 2025], following the reference pipeline of [Zhong et al., 2026, Liao et al., 2025].
- **Training:** IsaacLab [Mittal et al., 2023] with thousands of parallel environments, mixed terrain, and push/mass/friction randomisation.
- **Sim-to-sim:** MuJoCo evaluation as a transfer gate prior to hardware.
- **Hardware:** indoor calibrated trials and outdoor field demonstrations.

4.2 Performance Measures

- **Shooting (comparable to prior work):** shot error (nearest post-contact ball–target distance), success rate at 0.5 m and 1.0 m, peak post-contact ball speed, contact rate, alive rate.
- **Locomotion generality (not reported by prior soccer systems):** linear/angular command tracking error across the full command range including lateral and backward motion, terrain traversal success, push-recovery rate—measured *with the skill library attached*, to demonstrate the base is not degraded.
- **Transition quality:** post-skill fall rate, time-to-command-recovery after skill termination, and phase-resolved failure decomposition (fraction of failures occurring in approach, strike, and post-skill windows).
- **Skill-library coverage:** target-direction and ball-placement heat maps per skill and for the library, quantifying the region where N skills exceed single-skill coverage.

4.3 Current SOTA

Method	Regime	suc.@0.5m (%)	Shot err. (m)	$v_{\text{ball}}^{\text{max}}$ (m/s)	Alive (%)
PPO [Schulman et al., 2017]	Free-kick	0.0	4.721	0.780	0.0
AMP [Peng et al., 2021]	Free-kick	0.9	3.733	1.771	41.6
PAiD [Kong et al., 2026]	Free-kick	8.2	1.850	4.986	67.3
RoboNaldo [Zhong et al., 2026]	Free-kick	28.8	0.899	14.792	100.0
RoboNaldo [Zhong et al., 2026]	Moving	32.4	1.131	13.875	98.8

Table 2: Simulation SOTA for humanoid soccer shooting, as reported by Zhong et al. [2026]. On hardware, RoboNaldo attains 0.73 m free-kick error from 3 m with 91.2% contact and 13.10 m/s peak ball speed. Xu et al. [2026] (ICRA 2026) and Kong et al. [2026] additionally target egocentric perception. No system in this table reports locomotion-generality metrics.

Our target is to match shooting quality within the single-skill regime while demonstrating two capabilities absent from the table: arbitrary commanded locomotion with the skill library attached, and a multi-skill strike repertoire with measurably broader placement coverage.

5 Target Conference/Journal

- ICRA 2027 (submission approx. September 2026) — primary target; note Xu et al. [2026] appeared at ICRA 2026, so positioning against it is required.

Deadlines above are approximate and should be verified against official calls.

Timeline (from August 2026)

Aug 12 – Aug 25, 2026	Retarget $N=3-4$ additional kick clips (weak-foot, inside-foot pass, lateral flick); validate each in isolation on the existing single-skill stack.
Aug 26 – Sep 08, 2026	Train the multi-skill shared policy; ablate the skill-conditioning interface; verify no locomotion degradation with skills attached.
Sep 09 – Sep 22, 2026	Handoff ablation: scripted stabilisation tail vs. learned locomotion handoff, across all skills; phase-resolved failure decomposition.
Sep 23 – Oct 06, 2026	Locomotion-generalizability benchmark (command tracking, terrain, push recovery) and skill-coverage heat maps.
Oct 07 – Oct 27, 2026	Sim-to-sim gating in MuJoCo; hardware bring-up; indoor calibrated trials.
Oct 28 – Nov 17, 2026	Outdoor demonstration: commanded navigation interleaved with multi-skill strikes under a high-level controller.
Nov 18 – Dec 01, 2026	Baseline reproduction, final ablations, plots, and write-up.

Table 3: Work plan. The critical path is hardware validation; the multi-skill and handoff experiments are the two that carry the contribution.

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